

Proteins fold to reduce their energy

The shape into which a protein folds is determined by its amino acid sequence.

The chemical identity of the amino acid side chains influences the lowest-energy conformation of the peptide backbone, thus influencing the folded structure of the protein.

Primary Structure	The amino acid sequence.
Secondary Structure	Local folded structure. Stabilized by H-bonds between backbone amides.
Tertiary Structure	Interactions between regions of secondary structure. Stabilized by the hydrophobic effect and intermolecular interactions between side chains.
Quaternary Structure	Interactions between multiple folded polypeptide chains.

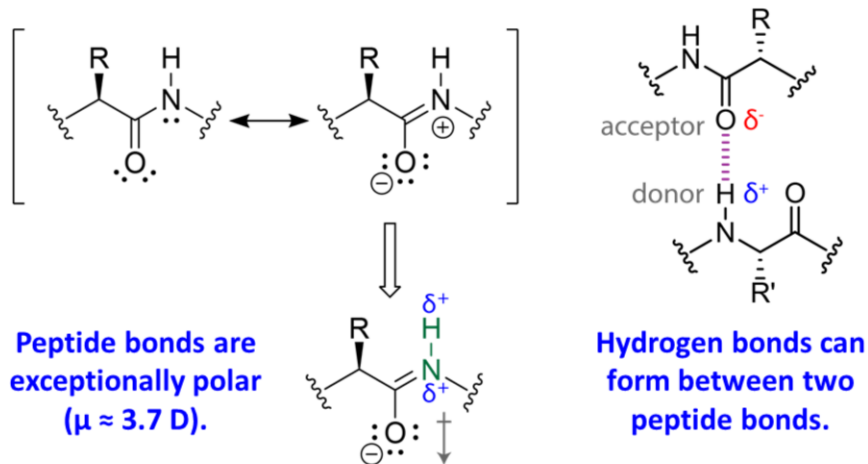
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Proteins are linear sequences of amino acids, but they do not remain extended in solution (or in cells). Instead, most proteins spontaneously fold into complex three-dimensional shapes. Protein chains fold in this way because it is energetically favorable to do so, and by folding they reduce their energy. Proteins fold on their own into specific three-dimensional shapes exclusively due to the information contained in their amino acid sequences. The chemical identities of the side chains determine the lowest-energy conformation of the polypeptide backbone, biasing the polypeptide to assume specific conformations, which causes the protein to fold into particular shapes.

Protein structure is divided into four “levels.” The most basic level of protein structure is primary structure, which is simply the linear sequence of amino acids in the polymer. Secondary structure describes local stretches of folded polymer. Secondary structure is stabilized by hydrogen bonds between peptide bonds. These hydrogen bonds between peptide bonds are the defining feature of secondary structure. Tertiary structure is the higher order three-dimensional folding that occurs when regions of secondary structure interact with each other. Tertiary structure is stabilized by interactions between side chains and by the hydrophobic effect (we’ll learn more about this later). Quaternary structure describes interactions between multiple folded polypeptide chains. On a chemical level quaternary structure is stabilized by the same forces that stabilize tertiary structure (i.e., intermolecular forces and the hydrophobic effect). Thus we can think of protein structure as nested, like little Russian dolls: secondary structure contains primary structure, tertiary structure contains secondary structure, and quaternary structure contains tertiary structure.

Hydrogen bonding between backbone amides is the basis of secondary structure



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The peptide bond is especially polar due to the resonance form in which positive and negative charges are found on the nitrogen and oxygen atoms, respectively. Consequently, both oxygen and nitrogen carry significant amounts of partial charge. Since the N-H bond is polar, the partial positive charge carried by nitrogen is spread out onto the hydrogen atom. The dipole moment of the peptide bond (3.7 D) is actually substantially larger than the rather large dipole moment of water (1.85 D).

The hydrogen atom in the peptide bond is a very good hydrogen bond donor since it carries a significant amount of positive charge. Similarly, the carbonyl oxygen in the peptide bond is a very good hydrogen bond acceptor since it carries a significant amount of negative charge. Hydrogen bonds between the oxygen and hydrogen atoms of two peptide bonds are the basis of secondary structure.